NammaVaani

A voice-first farming companion for Indian farmers

Speak naturally. Get practical farm help, with sources and confirmation.

THE PROBLEM

Farmers may need to move between separate services to check weather, look up market prices or report equipment problems. Text-heavy screens, unfamiliar terminology and language barriers can make these everyday tasks harder. A voice interface helps a farmer start with the way they already communicate: by speaking.

THE SOLUTION

NammaVaani is a responsive web prototype that turns a farmer's spoken request into a focused task. The farmer selects Kannada, Telugu or English, records a question, reviews the recognised text and receives a spoken response. When a required detail is missing, the application asks a follow-up instead of silently guessing.

WHAT A FARMER CAN DO

- Weather: request a forecast for a saved or specified location.
- Mandi prices: look up daily Agmarknet market observations, with a CSV fallback.
- Repair request: provide the equipment, issue and village; a ticket is saved after confirmation.
- Official letter: prepare a request for crop-protection guidance from an Agriculture Officer.
- Crop information: view the regional UAS Bengaluru Zone V crop basket.

A FARMER TASK IN PRACTICE

For example, a Telugu-speaking farmer reports a borewell motor problem. NammaVaani gathers the equipment, issue and village details, asks the farmer to confirm, then saves a repair ticket. It records the request; it does not claim that a technician has been dispatched.

HOW IT WORKS

1. Farmer speaks in a selected supported language.
2. Gnani Prisma v2.5 converts speech to text (STT).
3. NammaVaani routes the request to an application workflow, checks relevant data and asks for missing details when needed.
4. Gnani Timbre v2.5 converts the response to speech (TTS).
5. Farmer sees or hears the answer, draft or confirmed repair ticket.

WHY IT CAN BE TRUSTED

- Asks for missing details one question at a time rather than assuming.
- Shows source, location and retrieval time for weather and market information; market observations include their date.
- Requests consent before using saved GPS coordinates.
- Requires farmer confirmation before creating a repair ticket.
- Drafts an official request; it does not prescribe pesticides or claim that a technician has been dispatched.

GNANI AI INTEGRATION

Gnani Prisma v2.5 provides speech-to-text and Gnani Timbre v2.5 provides text-to-speech. NammaVaani supplies the farmer-focused intent handling, task flows, data lookups, clarifying questions and confirmation steps around those speech capabilities. A valid Gnani account key is required for live API calls.

LOCAL RELEVANCE

Built-in task replies support Kannada, Telugu and English, including phrase matching for common regional expressions and some English-mixed speech. A farmer profile can store state, district, taluk/mandal, hobli and village so supported services can use local context.

EVIDENCE AND HONEST LIMITS

The repository includes automated tests and an API verification script. The tests check application behaviour; a separate live API check is needed to confirm that the configured Gnani endpoints respond with the supplied account credentials. Full built-in task workflows are Kannada, Telugu and English; seven additional speech locales are beta and require an optional translation bridge. Prisma language auto-detection is not enabled by default. Direct Agmarknet access requires a data.gov.in API key; otherwise the CSV fallback is used. Mandi values are daily wholesale observations, not guaranteed real-time or farm-gate prices. The current prototype needs stronger multi-user authentication, privacy controls and expert review before broad public deployment.

Project repository: https://github.com/thanishraju6803-collab/NammaVaani

Live prototype: https://nammavaani.onrender.com